

Weekly Notice 14

Last exercise sessions: See WN 13.

Main questions for the exam:

1. Markov chains: finite-dimensional marginal distributions and existence.
2. Extensions of the Markov property.
3. Recurrence and transience.
4. Invariant measures and stationary distributions.
5. Lévy processes, regular convolution semigroups and the Markov property.
6. The strong Markov property for Lévy processes.
7. Brownian motion: existence and/or selected properties.
8. Properties of the Poisson process.
9. Wald's identities. Some of their applications and/or the first passage time τ_b for a Brownian motion.
10. Kolmogorov–Chentsov's theorem.

If question 4 is drawn, the presentation must involve (proofs and) results from Norris.

Curriculum:

1. All weekly notices. In particular, all exercises in the weekly notices.
2. Stochastic processes (Jan Pedersen): Chapters 1–14, Appendices A–H.
The following are not part of the curriculum: Section 9.5; Theorems 10.9 and 10.11; Example 11.19. Proposition 11.5 is cursory.
3. A Meticulous Introduction to Stochastic Calculus (Thomas Norman Dam, Mathias Mørck Ljungdahl, Anders Philbert Staudt Michelsen): Theorem 3.25 and Corollary 3.26.
4. Markov Chains (John Norris): Theorem 1.7.2, Theorem 1.7.5, Theorem 1.7.6 and Theorem 1.7.7.
5. Markov Jump Processes: Birth and Death Processes, Asymptotics (ChatGPT).

The exam:

1. The first student draws a question at 8:30.
2. At the exam, you will first present the chosen main question (approx. 12–15 minutes). We may ask questions along the way.
3. After the presentation, there will be a discussion of various questions related to the curriculum.